

PAPER_1324 — Solar Hale Cycle T = 22 yr EXACT

Author: Daniel T. Murphy **Framework:** UQFF — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology Open (CLOSED) **Date:** June 11, 2026 **Calculator:** `calculate_paradox({"paradox": "solar_dynamo"})`

Master Expression

$T_{\text{Hale}} = D_{\text{crit}} - D_{\text{phys}} = 26 - 4 = 22 \text{ yr EXACT}$

Match: 0.000% EXACT

Interpretation

Solar dynamo Hale cycle is an integer-primitive identity. Sunspot $11\text{-yr} \times 2 = 22\text{-yr}$ full polarity flip.

UQFF Locked Primitives

- $D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, SO_5 = 10, A_5 = 60, N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12, \beta_i = 0.6029, \Phi_{\text{res}} = 0.84, \Lambda = 0.00729735, F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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